

solution to method to the mathmess

the puzzle is a mono-alphabetic substitution cipher: all symbols decrypt to another symbol present in the puzzle. analyzing the positioning and distribution of various symbols (many, many lines of reasoning possible; see following pages), the ciphertext decrypts to the following plaintext:

$$0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < \infty$$

$$x! = \prod_{n=1}^x n$$

$$\zeta(x) = \sum_{n=1}^{\infty} n^{-x}$$

$$\pi = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-f(x,y)} dx dy$$

$$f(x,y) = x^2 + y^2$$

$$\nabla f(x,y) = (2x, 2y)$$

$$A(1) = \sum_{n=0}^2 f(n, n+1)$$

$$A(2) = \prod_{n=1}^2 f(n, n)$$

$$A(3) + A(1) = 4!$$

$$A(4) = \nabla f(6, 3/2) \cdot \nabla f(1, -7/2)$$

$$A(5) = 3 \int_{-1}^1 \int_{-1}^1 f(x,y) dx dy - e^{\pi(-1)^{1/2}}$$

$$A(6) = \pi^2 / \zeta(2)$$

$$A(7) = f(A(4), 0)$$

$$A(8)! = A(3!)$$

evaluating each $A(i)$ and indexing gives the answer of **SPECIFIC**:

$$\begin{array}{l|l} A(1) = 19 & \text{S} \\ A(2) = 16 & \text{P} \\ A(3) = 5 & \text{E} \\ A(4) = 3 & \text{C} \\ A(5) = 9 & \text{I} \\ A(6) = 6 & \text{F} \\ A(7) = 9 & \text{I} \\ A(8) = 3 & \text{C} \end{array}$$

below is an example of a possible line of reasoning (ciphertext in red, plaintext in blue).

- $\overset{-}{7}_{=-}$ $\overset{-}{7}_{=-}$ and $\overset{0}{7}_{=0}$ $\overset{0}{7}_{=0}$ clearly suggest that $\boxed{\overset{0}{7} = \int}$.
 - yey/ appears after both integral expressions (the expressions with $\overset{0}{7} = \int$), suggesting that $\boxed{\text{yey/} = dx dy}$.
- $\overset{e}{\int}_{\zeta \nabla 0}$, $\overset{-}{\int}_{\zeta \nabla 0}$, $\overset{x}{\int}_{\zeta \nabla !}$, and $\overset{x}{\int}_{\zeta \nabla 0}$ suggest that $\int$ and $\int$ are $\sum$ and $\prod$, though we can't yet tell which is which.
 - $\zeta \nabla$ appears at the bottom of each of these, suggesting that $\boxed{\zeta \nabla = n =}$.
- $!A0AxAfA\infty A \prod A \cdot A\pi A4A-$ is quite conspicuous due to the chain of A 's; one might guess that $A = <$, which means this line decodes as follows (note that there is no 9 in the puzzle):
 $\boxed{!A0AxAfA\infty A \prod A \cdot A\pi A4A- = 0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < \infty}$
 - (one might also guess that $A = +$ or $-$, but these guesses don't allow the decryption of the rest of this line to fall into place so nicely.)

at this point we have:

$$\begin{aligned}
 &0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < \infty \\
 &x1 = \int_{n=1}^x n \\
 &) < x3 = \int_{n=1}^{\infty} n^{=x} \\
 &n = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sum^{=6 < x} (y^3 dx dy \\
 &6 < x(y3 = x^2 5y^2 \\
 &+6 < x(y3 = < 2x(2y3 \\
 &8 < 13 = \int_{n=0}^2 6 < n(n513 \\
 &8 < 23 = \int_{n=1}^2 6 < n(n3 \\
 &8 < 3358 < 13 = 41 \\
 &8 < 43 = + 6 < 6(3223d + 6 < 1 (= 7223 \\
 &8 < 53 = 3 \int_{=1}^1 \int_{=1}^1 6 < x(y3 dx dy = \sum^{n < = 13^{122}} \\
 &8 < 63 = n^2 2) < 23 \\
 &8 < 73 = 6 < 8 < 43(03 \\
 &8 < 831 = 8 < 313
 \end{aligned}$$

(i'm sure you don't envy me typesetting that.) now:

4. $\int_{=\infty}^{\infty} \int_{=\infty}^{\infty}$ and $\int_{=1}^1 \int_{=1}^1$ suggest that $\boxed{= = -}$.

5. $< * 3$ appears frequently—i.e. < 3 frequently surrounds one or a few symbols—suggesting that $\boxed{< 3 = ()}$.

6. recall step 2. looking at $x1 = \int_{n=1}^x n$, $\boxed{\int = \prod}$ is nice so that $\boxed{1 = !}$. then by process of elimination, $\boxed{, = \sum}$.

7. combining steps 4, 5, and 6, $, < x3 = \int_{n=1}^{\infty} n^{-x}$ decodes to $((x) = \sum_{n=1}^{\infty} n^{-x}$, meaning $\boxed{(= \zeta}$ (this is the definition of the Riemann zeta function).

now we have:

$$0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < \infty$$

$$x! = \prod_{n=1}^x n$$

$$\zeta(x) = \sum_{n=1}^{\infty} n^{-x}$$

$$n = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \sum^{-6(x(y))} dx dy$$

$$6(x(y)) = x^2 5y^2$$

$$+6(x(y)) = (2x(2y))$$

$$8(1) = \sum_{n=0}^2 6(n(51))$$

$$8(2) = \prod_{n=1}^2 6(n(n))$$

$$8(3) + 8(1) = 4!$$

$$8(4) = +6(6(322))d + 6(1(-722))$$

$$8(5) = 3 \int_{-1}^1 \int_{-1}^1 6(x(y)) dx dy - \sum^{n(-1)} 1^{22}$$

$$8(6) = n^2 2\zeta(2)$$

$$8(7) = 6(8(4))(0)$$

$$8(8)! = 8(3!)$$

8. $($ appears frequently between two variables, suggesting that $\boxed{(= ,}$.

9. 6 and 8 always appear before $($, suggesting that they are functions. f and A are the only remaining sensible options for functions (π is iffy and will be conclusively eliminated in step 11). say $\boxed{6 = f}$ and $\boxed{8 = A}$ (the puzzle also works the other way around, but i used A for “answer”).

omitting the first three lines (which are already fully decoded), we now have

$$\begin{aligned}
 n &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Sigma^{-f(x,y)} dx dy \\
 f(x,y) &= x^2 5 y^2 \\
 +6(x,y) &= (2x, 2y) \\
 A(1) &= \sum_{n=0}^2 f(n, n51) \\
 A(2) &= \prod_{n=1}^2 f(n, n) \\
 A(3) + A(1) &= 4! \\
 A(4) &= +6(6, 322)d + 6(1, -722) \\
 A(5) &= 3 \int_{-1}^1 \int_{-1}^1 f(x,y) dx dy - \Sigma^{n(-1)^{122}} \\
 A(6) &= n^2 2 \zeta(2) \\
 A(7) &= f(A(4), 0) \\
 A(8)! &= A(3!)
 \end{aligned}$$

10. looking at $f(x,y) = x^2 5 y^2$ and $A(1) = \sum_{n=0}^2 f(n, n51)$, the only remaining symbols that really make sense in 5 's place are $+$ and $/$, but $5 = /$ would give $n51 = n/1$, which would be verbose/unnatural; instead, letting $\boxed{5 = +}$ cleanly implies $\boxed{+ = \nabla}$, since $+f(x,y) = (2x, 2y)$.
11. looking at $n = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Sigma^{-f(x,y)} dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Sigma^{-(x^2+y^2)} dx dy$, you might want $\boxed{\Sigma = e}$, so that the integral equals $\boxed{n = \pi}$ (this is a well-known intermediate result in calculating the [Gaussian integral](#)).
12. applying step 11 to $A(5) = \dots \Sigma^{n(-1)^{122}}$ gives $\dots e^{\pi(-1)^{122}}$, which resembles the familiar expression $e^{\pi i}$; this indeed works, if $\boxed{2 = /}$.
13. by process of elimination (and by context), $\boxed{d = \cdot}$.